

# Information-State-Based Reinforcement Learning for the Control of Partially Observed Nonlinear Systems

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**Abstract**—This article develops a model-based reinforcement learning (RL) approach to the closed-loop control of nonlinear dynamical systems with a partial nonlinear observation model. We propose an “information-state”-based approach to rigorously transform the partially observed problem into a fully observed problem where the information state consists of the past several observations and control inputs. We further show the equivalence of the transformed and the initial partially observed optimal control problems and provide the conditions to solve for the deterministic optimal solution. We develop a data-based generalization of the iterative linear quadratic regulator (ILQR) for the RL of partially observed systems using a local linear time-varying model of the information-state dynamics approximated by an autoregressive-moving-average (ARMA) model that is generated using only the input–output data. This approach allows us to design a local perturbation feedback control law that provides an optimum solution to the partially observed feedback design problem locally. The efficacy of the developed method is shown by controlling complex high-dimensional nonlinear dynamical systems in the presence of model and sensing uncertainty.

**Index Terms**—Nonlinear systems, optimal control, partial differential equation (PDE) control, partial-state observation, reinforcement learning (RL), robotics.

## I. INTRODUCTION

THE problem of optimal control for nonlinear dynamical systems with partial observation finds several applications and is still considered an unsolved problem [1], [2]. Computing the optimal feedback law in the entire domain is computationally intractable for complex high-order systems due to the “curse of dimensionality” associated with solving dynamic programming (DP) [1]. The problem becomes more challenging when we have access to only a black box model, of which only some of the states of the system are available for measurement, i.e., under partial-state observation, which tends to be the case for most of the problems. In control system theory, methods that use only the outputs of the system (as opposed to the full state) for control are referred to as

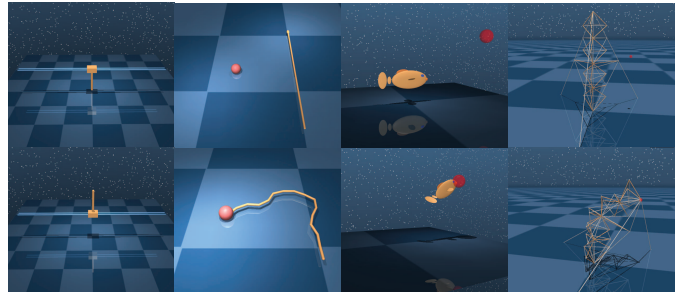


Fig. 1. Models simulated in MuJoCo in their initial and final states. In this article, we assume that we only have access to partially observed data from computational models of the systems shown above.

output feedback [3] methods. The reinforcement learning (RL) community refers to it as a partially observed problem or a partially observed Markov decision process (POMDP).

In this work, we propose a model-based on-policy RL approach for learning to optimally control complex partially observed nonlinear dynamical systems. The approach presented is model-based in the sense that we find an approximation of the system model from input–output data to calculate the optimal feedback rather than directly learning the feedback law from data, as done in model-free RL. The primary idea is to convert the partially observed optimal control problem to a “fully observed” problem described using the *information state* [1] and to show that the problem described using the information state is equivalent to the original problem. Then, we use a data-driven approach to solve the information state optimal control problem. Like in other RL applications, we have access to a simulator which we use to perform rollouts and collect input–output data. We use this input–output data to improve our policy and ultimately arrive at the optimum law.

The proposed method is applicable to complex robotic systems such as those shown in Fig. 1. Complex systems like the tensegrity robotic arm and fish robot interacting with the fluid medium lack analytical models, and the full state of the system is typically not measured. Soft robots also have similar challenges and need a data-based solution [4]. Similar techniques are also used to control partial differential equations (PDEs) [5], and we show an example in the experiments section to control PDEs that are partially observed. It can also

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be used in data-driven control design for autonomous vehicles [6] and other robotics applications.

One approach to solving partially observed problems is to use a feedback policy that acts on the current output, known as static output feedback control [7]. However, static output feedback approaches are suboptimal compared to full-state feedback control [8]. Thus, traditional approaches use a nonlinear observer (such as a high-gain observer, extended Kalman filter, or particle filter) to reconstruct the full state of the system, which is then used for the full-state feedback control [7], [9]. Unfortunately, observer-based approaches and other nonlinear control techniques such as output feedback linearization cannot be utilized in the problems studied in this article since an analytical model is unknown [1] (see Fig. 1). Adaptive control-based approaches are sample efficient and near optimal and can handle uncertainties, but they are predominantly applicable only for linear systems and regulation applications [10], [11] and rarely applicable for the class of nonlinear systems considered in this article.

There has been a significant body of work in *learning to control* unknown dynamical systems through approximate DP (ADP) and RL techniques [12], [13], [14]. These approaches have been successfully used in many areas like playing games [13], locomotion [15], robotic hand manipulation [16], and autonomous vehicles [6]. However, despite excellent performance on several tasks, model-free RL is still considered very data-intensive, with typically a really large training time. They also require meticulous hyperparameter tuning, without which they face convergence issues [17]. Both of these challenges severely limit the applicability of such methods to complex, real-world domains. Solving partially observable problems in RL is significantly harder due to several core challenges that arise from the incomplete knowledge of the system's state. Typically, one uses the history of observations and actions or the information state as a surrogate for the system's state, but this increases the computational complexity as the agent has to learn in a high-dimensional information space. Most approaches that consider information state to solve partially observed problems are limited to either linear system models [3], [18], [19] or finite state and action spaces [20]. Works that consider continuous state-action space use a memory-based element like a recurrent neural network (RNN), predominantly a long short-term-memory (LSTM) network, to model the hidden state of the system [21], [22], [23]. Although the approaches empirically perform well on benchmarks, there is only limited work on formal analysis [24] of the partially observed problem and for finite state, control, and observation space problems. Consequently, they are inapplicable to the nonlinear continuous state, control, and observation space problems, unlike the information state perturbation feedback RL approach proposed in this article.

#### A. Contributions

The proposed approach is a generalization of the so-called decoupled data-based control (D2C) [25], [26], [27], a model-based RL approach, to partially observed systems. This article builds on the work in [28] as follows: 1) we provide the theoretical justification of the information-state approach to

partially observed problems (Section III); 2) we utilize a minimum principle (MP) in the information state (Section IV) to show that satisfying the MP is sufficient for a globally optimum open-loop solution (Section IV-A) and derive the perturbation terms in the optimal feedback law (Section IV-B); 3) we generalize the iterative linear quadratic regulator (ILQR) approach [29], [30] to partially observed problems in a model-based RL fashion using a linear time varying autoregressive moving average (LTV-ARMA) model identification approach to solve the open-loop optimal control problem along with the linear perturbation feedback (Section V); and 4) we generate a heuristic closed-loop feedback design under noise by augmenting the feedback law with a Kalman filter in the information state (Section VI) that is shown to be capable of handling significant levels of process and sensor noise in complex nonlinear partially observed problems. To the best of our knowledge, all of the contributions above are novel and constitute a systematic and theoretically justified procedure for model-based RL in nonlinear partially observed problems.

## II. PROBLEM FORMULATION

Consider a nonlinear discrete-time dynamical system

$$x_{k+1} = f(x_k) + g(x_k)u_k \quad (1a)$$

$$z_k = h(x_k) \quad (1b)$$

where  $x_k \in \mathbb{R}^{n_x}$  is the state;  $u_k \in \mathbb{R}^{n_u}$  is the control input; and  $z_k \in \mathbb{R}^{n_z}$  is the output of the system defined  $\forall k \geq 0$ , with  $f(0) = 0$  and  $g(0) \neq 0$ . The objective of this work is to find the optimal control inputs  $\{u_0, u_1, \dots, u_{N-1}\}$  that minimize the cost

$$J(x_0) = \sum_{k=0}^{N-1} c(z_k, u_k) + c_N(z_N) \quad (2)$$

subject to the system model in (1), where  $c(z_k, u_k)$  denotes a running incremental cost and  $c_N(z_N)$  denotes a terminal cost function. The goal is to find an output feedback policy, i.e., a policy that only has access to the observations  $z_k$ , such that the cost above is minimized from an unobserved initial state  $x_0$ .

*Note:* We use the term partially observed to mean that the full state is not available for measurement. We consider the measurements used in the training phase to be noise-free, but we consider process and measurement noise when we test the control policy.

#### A. Heuristic Overview and Structure of Manuscript

We give an overview of the structure of this article and a heuristic summary of the results to aid the reader in understanding the big picture before delving into the details. The broad structure is as follows. In Sections III and IV, we consider the theoretical properties of the partially observed problem posed above. Then, in Sections V and VI, we use the theoretical results to develop a model-based approach to RL in such systems. The particulars of the developments in the different sections are outlined as follows.

In Section III, we introduce the information state  $\mathcal{Z}_k = [z_k^T, z_{k-1}^T, \dots, z_{k-q}^T, u_{k-1}^T, \dots, u_{k-q}^T]^T$ , which consists of the past  $q$  observations and control inputs. We show that there

exists a unique map that relates the state  $x_k$  and the information state  $\mathcal{Z}_k$  assuming observability. Then, we show that the original partially observed state-space problem can be posed as a fully observed problem in the information state and show their equivalence.

Section IV deals with the necessary and sufficient conditions for the optimal control problem. Under the assumption that the optimal cost-to-go function is continuously differentiable, we show that the MP is also sufficient for obtaining the unique global optimum. In Section IV-B, we perform a perturbation expansion of the cost-to-go as follows  $J_k(\mathcal{Z}_k) = \bar{J}_k + \bar{G}_k^\top \delta \mathcal{Z}_k + (1/2) \delta \mathcal{Z}_k^\top \bar{P}_k \delta \mathcal{Z}_k + \dots$ , where  $\delta \mathcal{Z}_k = \mathcal{Z}_k - \bar{\mathcal{Z}}_k$  is the perturbation of the information state from the optimal nominal open-loop information state trajectory  $\bar{\mathcal{Z}}_k$ , and derive the equations that govern the terms in the expansion. This allows us to obtain the optimal perturbation feedback  $u_k = \bar{u}_k + K_k \delta \mathcal{Z}_k + \dots$ .

Section V outlines an information state-based ILQR procedure inspired by the perturbation feedback design aimed at computing the optimal linear perturbation feedback law  $u_k = \bar{u}_k + K_k \delta \mathcal{Z}_k$ . The key step in this algorithm is identifying the linear time-varying system around a trajectory from input-output data. We use the LTV system identification using an autoregressive moving average model proposed in [31] and use this model to compute the local correction and iteratively improve the linear perturbation feedback until it converges to the optimum.

In Section VI, we explain how we account for uncertainty in the form of process and measurement noise. In essence, we use the linear perturbation feedback design from Section V and design a Kalman filter for the local time-varying system around the trajectory to account for the process and sensor noise. Although a theoretical treatment of this procedure is beyond the scope of this article, computational results presented in Section VII indicate that this design successfully manages significant levels of noise.

### III. INFORMATION-STATE-BASED CONTROL PROBLEM

In this section, we introduce the information state and discuss how to transform the original optimal control problem in (2) to the information-state domain.

Let  $f^n(x_{k-q}; u_{k-q}, u_{k-q+1}, \dots, u_{k-q+n-1})$  denote the map from the state at time  $k-q$ ,  $x_{k-q}$ , to the state  $x_{k-q+n}$  at time  $k-q+n$ . Given the initial state and inputs from time  $k-q$  to time  $k-1$ , i.e.,  $\{x_{k-q}; u_{k-q}, u_{k-q+1}, \dots, u_{k-1}\}$ , we can write the following expressions for the observations  $\{z_{k-q}, z_{k-q+1}, \dots, z_k\}$  as:

$$\begin{aligned} z_{k-q} &= h(x_{k-q}) \\ z_{k-q+1} &= h(f^1(x_{k-q}; u_{k-q})) \\ &\vdots \\ z_k &= h(f^q(x_{k-q}; u_{k-q}, u_{k-q+1}, \dots, u_{k-1})). \end{aligned} \quad (3)$$

In the partially observed problem, one can find the underlying state  $x_{k-q}$ , which is the solution to the above set of nonlinear equations. Let  $Z_k^q = [z_{k-q}^\top, z_{k-q+1}^\top, \dots, z_k^\top]^\top$ ,  $U_k^q = [u_{k-q}^\top, u_{k-q+1}^\top, \dots, u_{k-1}^\top]^\top$ , and  $H_k^q(x_{k-q}) = [h(x_{k-q})^\top, \dots, h(f^q(x_{k-q}; u_{k-q}, \dots, u_{k-1}))^\top]^\top$ .

*Assumption 1: Observability:* Assume that there exists a finite  $\bar{q}$ , such that for all  $q \geq \bar{q}$ , (3) has a unique solution for  $x_{k-q}$ , regardless of  $(Z_k^q, U_k^q)$ .

This assumption essentially means that the initial state can be unambiguously reconstructed from a finite history of measurements and control inputs. However, note that this condition needs only to be satisfied in principle, and we do not ever explicitly try to solve this equation in our control synthesis. We do not need the knowledge of the functions  $f^n(\cdot), h(\cdot), H_k^q(\cdot)$ . We use only input-output data to find the control policy. For the rest of this article, we consider  $q$  used to satisfy  $q \geq \bar{q}$ .

Due to the implicit function theorem [32] and the observability assumption, we can write  $x_{k-q}$  as some unique function  $\bar{f}(\cdot, \cdot)$  of past measurements and inputs as

$$x_{k-q} = \bar{f}(Z_k^q, U_k^q). \quad (4)$$

*Remark 1:* Note that typically for a partially observed problem, one would use a (nonlinear) observer to estimate the state, which would then be used to specify the control action, assuming the estimated state to be the true state (certainty equivalence). Owing to the observability assumption 1, given the first  $\bar{q}$  inputs and outputs, one can, in principle (if the model is known), exactly reconstruct the initial state and thus predict the state evolution exactly after  $\bar{q}$  steps, thereby mapping back to the fully observed problem. Therefore, we shall modify the original formulation and assume that the first  $\bar{q}$  inputs are specified in the partially observed control formulation we discuss later in this section. Finally, note that a typical nonlinear observer can never perfectly reconstruct the state in a finite number of steps.

Next, let us write the state at the current time  $x_k$  as  $f^q(x_{k-q}; u_{k-q}, u_{k-q+1}, \dots, u_{k-1})$ , which can be written again by substituting for  $x_{k-q}$  from (4) in some unique functional form, based on observability assumption and implicit function theorem, as  $x_k = \Psi(Z_k^q, U_k^q)$ .

Let us now finally define the “information state.”

*Definition 1 (Information State):* The information state [1], [33] of the system in (1) of order “ $q$ ” (at time  $k$ ), is defined as  $Z_k^q = [z_k^\top, z_{k-1}^\top, \dots, z_{k-q}^\top, u_{k-1}^\top, \dots, u_{k-q}^\top]^\top \in \mathbb{R}^n$ , where  $n = (q+1)n_z + qn_u$ .

The above definition allows us to write:  $x_k = \Psi(Z_k^q)$ . Furthermore, due to the implicit function theorem, if the dynamics and the observation functions  $f$ ,  $g$ , and  $h$  are  $\mathcal{C}^k$ , so is the map  $\Psi$ . The above development can be summarized as follows.

*Lemma 1:* Given Assumption 1, there exists a unique function  $\Psi(\cdot)$  such that the state at time  $k$ ,  $x_k = \Psi(Z_k^q)$ . In particular, if  $f(\cdot)$  and  $h(\cdot)$  are  $\mathcal{C}^k$ , so is the function  $\Psi$ .

The above result shows that the state at time  $k$  is some nonlinear map of the observation and the control inputs at the previous “ $q$ ” time steps.

The remainder of this section shows how to transform the optimal control problem into the information-state domain. Consider the observation of the system as  $z_k = h(x_k) = h(f(x_{k-1}) + g(x_{k-1})u_{k-1})$ . Using the relationship,  $x_{k-1} = \Psi(Z_{k-1}^q)$ , the observation can be written in the form,  $z_k = \bar{h}(Z_{k-1}^q, u_{k-1})$ , and furthermore, the equation can be



written in terms of information state as  $\mathcal{Z}_k^q = \mathcal{H}(\mathcal{Z}_{k-1}^q, u_{k-1})$ . This can be summarized in the following result.

**Lemma 2:** Under Assumption 1, the systems  $x_{k+1} = f(x_k) + g(x_k)u_k$ ,  $z_k = h(x_k)$  and  $\mathcal{Z}_{k+1}^q = \mathcal{H}(\mathcal{Z}_k^q, u_k)$  have the same input–output response after  $k \geq q$ , regardless of the “unknown” initial condition  $x_0$ .

Recall that  $\bar{q}$  is the minimum number of observations required to satisfy Assumption 1. The information state model needs at least  $\bar{q}$  measurements and inputs before it can start prediction, as these are required to form the initial information state. We assume that the first  $\bar{q}$  control inputs and resulting observations are specified to us, which implies that the state  $x_{\bar{q}}$  is known given the observability assumption. Then, we may reformulate the original partially observed control problem as the following fully observed problem starting at the initial state  $x_{\bar{q}}$ :

$$J_x^* = \min_{u_k} \sum_{k=\bar{q}}^{N-1} c(z_k, u_k) + c_N(z_N) \quad (5a)$$

$$\text{s.t. } x_{k+1} = f(x_k) + g(x_k)u_k, \text{ given } x_{\bar{q}} \\ z_k = h(x_k). \quad (5b)$$

Furthermore, we pose the following “fully observed” optimal control problem in terms of the information state, assuming that the same initial  $\bar{q}$  inputs are applied as above

$$J_{\mathcal{Z}}^* = \min_{u_k} \sum_{k=\bar{q}}^{N-1} c(z_k, u_k) + c_N(z_N) \quad (6a)$$

$$\text{s.t. } \mathcal{Z}_{k+1}^q = \mathcal{H}(\mathcal{Z}_k^q, u_k), \text{ given } \mathcal{Z}_{\bar{q}}^q. \quad (6b)$$

Hence, we have transformed our optimal control problem to the information-state domain in (6).

#### A. Optimality of the Information-State Control Problem

This section discusses the equivalence of solving the optimal control problem using the information-state dynamics and the partially observed state-space dynamics and proves that transforming the problem to the information state leads to no loss in optimality after a finite initial transience. The equivalence of the solution to the two problems is established in the following result.

**Theorem 1:** Given the initial information state  $\mathcal{Z}_{\bar{q}}^q$  or, equivalently, the state  $x_{\bar{q}}$ , the solution to the optimal control problem (5) is identical to the solution to the optimal information state control problem (6).

*Proof:* See Appendix A.  $\square$

**Remark 2:** Notice that the two problems (5) and (6) mention the initial state to be defined at  $k = \bar{q}$  and assume that the initial few control inputs  $\{u_0, u_1, \dots, u_{\bar{q}-1}\}$  are specified. Intuitively, one can see that these inputs are ambiguous as they are needed to get enough information to reconstruct the initial state/form the initial information state and should not be included in the optimization. Thus, it is advisable to take small perturbations to keep the system near the initial state till the state can be reconstructed at  $k = \bar{q}$ . Nonetheless, given that  $\bar{q} \ll T$ , we can expect that this initial transient will not affect the total cost significantly.

#### IV. OPTIMAL CONTROL OF THE INFORMATION STATE PROBLEM

In this section, we study the information-state optimal control problem. We make the following assumption for the problem given in (6) and show that the information-state dynamics is control affine.

**Assumption 2:** The incremental cost in (6) is quadratic in the control:  $c(z_k, u_k) = l(z_k) + (1/2)u_k^\top R u_k$ , where  $l(z_k) \geq 0$  and  $R$  is positive definite.

The systems we consider (Fig. 1) are continuous-time dynamical systems whose evolution is governed by ordinary differential equations (ODEs), and a large class of these systems have a control-affine structure. However, the measurements/outputs of the systems  $z_k$  are discrete with a sampling time of  $\Delta t$ . In general, the discrete-time information-state model would not have a control-affine structure unless the sampling time is sufficiently small, as shown in the following result.

**Proposition 1:** Let the dynamics of the underlying system be affine in control of the form:  $\dot{x} = f(x) + g(x)u$ . Then, if the sampling time  $\Delta t$  is sufficiently small such that  $O(\Delta t^2) \rightarrow 0$ , the information-state model is also affine in control

$$\mathcal{Z}_k^q = \mathcal{F}(\mathcal{Z}_{k-1}^q) + \mathcal{G}(\mathcal{Z}_{k-1}^q)u_{k-1}. \quad (7)$$

*Proof:* See Appendix B.  $\square$

Using the above results and considering the time index  $k = 0$  to be the first time index, the information state is constructed, and the problem in (6) can be rewritten as

$$J = \min_{u_k} \sum_{k=0}^{N-1} \left( l(z_k) + \frac{1}{2}u_k^\top R u_k \right) + c_N(z_N) \quad (8a)$$

$$\text{s.t. } \mathcal{Z}_{k+1} = \mathcal{F}(\mathcal{Z}_k) + \mathcal{G}(\mathcal{Z}_k)u_k, \text{ given } \mathcal{Z}_0. \quad (8b)$$

Superscript  $q$  in  $\mathcal{Z}^q$  and subscript  $\mathcal{Z}$  in  $J_{\mathcal{Z}}$  are ignored for the sake of convenience. The following result gives the necessary conditions for the solution  $\{\bar{\mathcal{Z}}_k, \bar{u}_k\}$  to the above fully observed problem in (8) that has to satisfy the MP in discrete time.

**Theorem 2: Necessary Conditions [7]:** Let the cost functions  $l(\cdot)$  and  $c_N(\cdot)$ , the drift  $\mathcal{F}(\cdot)$ , and the input influence function  $\mathcal{G}(\cdot)$  be  $\mathcal{C}^2$ , i.e., twice continuously differentiable. The minimum of the open-loop problem (8) starting at some initial information state  $\mathcal{Z}_0$  satisfies

$$\bar{u}_k = -R^{-1} \mathcal{B}_k^\top \bar{G}_{k+1} \quad (9)$$

$$\bar{\mathcal{Z}}_{k+1} = \mathcal{F}(\bar{\mathcal{Z}}_k) - \mathcal{G}(\bar{\mathcal{Z}}_k) R^{-1} \mathcal{G}(\bar{\mathcal{Z}}_k)^\top \bar{G}_{k+1} \quad (10)$$

$$\bar{G}_k = \bar{I}_k^{\mathcal{Z}} + \mathcal{A}_k^\top \bar{G}_{k+1} \quad (11)$$

where  $\mathcal{A}_k = \bar{\mathcal{F}}_k^{\mathcal{Z}} + \sum_{j=1}^{n_u} \bar{\Gamma}_k^{j,\mathcal{Z}} \bar{u}_k^j$ ,  $\mathcal{B}_k = \mathcal{G}(\bar{\mathcal{Z}}_k)$ , and  $\{\bar{\mathcal{Z}}_k\}$  represents the optimal nominal trajectory.  $\bar{I}_k^{\mathcal{Z}} = \nabla_{\mathcal{Z}} l|_{\bar{\mathcal{Z}}_k}$  and  $\bar{G}_k = \nabla_{\mathcal{Z}} J_k|_{\bar{\mathcal{Z}}_k}$ , with terminal condition  $\bar{G}_N = \nabla_{\mathcal{Z}} c_N|_{\bar{\mathcal{Z}}_N}$ , are column vectors.  $\bar{u}_k = [\bar{u}_k^1 \dots \bar{u}_k^{n_u}]^\top$ , the control influence matrix:  $\mathcal{G} = [\Gamma^1(\mathcal{Z}) \dots \Gamma^{n_u}(\mathcal{Z})]$ , and  $\Gamma^j$  and  $\bar{u}_k^j$  represent the control influence vector and optimal control vector corresponding to the  $j$ th input, respectively. Finally,  $\bar{\mathcal{F}}_k^{\mathcal{Z}} = \nabla_{\mathcal{Z}} \mathcal{F}|_{\bar{\mathcal{Z}}_k}$  and  $\bar{\Gamma}_k^{j,\mathcal{Z}} = \nabla_{\mathcal{Z}} \Gamma^j|_{\bar{\mathcal{Z}}_k}$  give the Jacobians of the system dynamics along the nominal trajectory.

### A. Sufficiency of the MP for a Global Optimum

We proceed to show that the problem in (8) has a unique minimum under some assumptions. The problem in (8) can equivalently be written as the following DP problem using Bellman's principle of optimality:

$$J_k(\mathcal{Z}_k) = \min_{u_k} \left[ l(z_k) + \frac{1}{2} u_k^\top R u_k + J_{k+1}(\mathcal{Z}_{k+1}) \right] \quad (12)$$

where  $J_N(\mathcal{Z}_N) = c_N(z_N)$ .

The DP equation (12), in some cases, might not have a smooth solution, i.e.,  $J_k(\cdot)$  might not be differentiable. However, we shall consider the case where it has a smooth solution and make the following assumption for the rest of our work.

*Assumption 3:* There exists a continuously differentiable function ( $\mathcal{C}^1$ ) that satisfies the DP equation (12).

For the sake of clarity and continuity, we restate some of the equations as follows. Let the state at time  $k$  be  $\mathcal{Z}_k$ , and then, the minimizing control using the first-order necessary condition is given by

$$u_k = -R^{-1} \mathcal{G}(\mathcal{Z}_k)^\top \frac{\partial J_{k+1}}{\partial \mathcal{Z}_{k+1}} \Big|_{\mathcal{Z}_{k+1}} = -R^{-1} \mathcal{G}(\mathcal{Z}_k)^\top G_{k+1}.$$

The state evolution using the optimal control is given by

$$\mathcal{Z}_{k+1} = \mathcal{F}(\mathcal{Z}_k) - \mathcal{G}(\mathcal{Z}_k) R^{-1} \mathcal{G}(\mathcal{Z}_k)^\top G_{k+1}. \quad (13)$$

Substituting this control in (12) gives the evolution of the optimal cost-to-go

$$J_k(\mathcal{Z}_k) = l(z_k) + \frac{1}{2} G_{k+1}^\top \mathcal{G}(\mathcal{Z}_k) R^{-1} \mathcal{G}(\mathcal{Z}_k)^\top G_{k+1} + J_{k+1}(\mathcal{F}(\mathcal{Z}_k) - \mathcal{G}(\mathcal{Z}_k) R^{-1} \mathcal{G}(\mathcal{Z}_k)^\top G_{k+1}). \quad (14)$$

Taking the partial derivative of (14) with respect to  $\mathcal{Z}_k$  gives rise to the costate equation

$$G_k = l_k^{\mathcal{Z}} + (\mathcal{F}_k^{\mathcal{Z}} - \nabla_{\mathcal{Z}} \mathcal{G} |_{\mathcal{Z}_k} R^{-1} \mathcal{G}(\mathcal{Z}_k)^\top G_{k+1})^\top G_{k+1}. \quad (15)$$

(Note: in (15),  $\nabla_{\mathcal{Z}} \mathcal{G}$  would be a tensor for the vector case, but we abuse the notation for the sake of illustration.) Since (12) is a terminal value problem, we start from the terminal state. For a given terminal condition  $\{\mathcal{Z}_N, G_N = (\partial c_N / \partial \mathcal{Z}_N) |_{\mathcal{Z}_N}\}$ , (13) and (15) can be solved backward in time to find the optimal solution. Hence, (13) and (15) are the characteristic equations to the solution to the DP problem in (12). However, in general, they need not have a unique backward/forward evolution. See Fig. 2 for an illustration of how the state–costate pair evolves.

*Remark 3:* Equations (13) and (15) are the discrete-time analog to the characteristic ODEs one obtained for the Hamilton–Jacobi–Bellman (HJB) equation using the method of characteristics [34]. In continuous time, the HJB equation for minimizing the cost  $J^u(x, 0) = \int_{t=0}^T (l(x_t) + (1/2) r u_t^2) dt + c_T(x_T)$  for the system  $\dot{x} = f(x) + g(x)u$  is given by

$$\begin{aligned} \frac{\partial J(x, t)}{\partial t} &= - \min_{u_t} \left[ l(x) + \frac{1}{2} r u_t^2 + \frac{\partial J(x, t)}{\partial x} (f(x) + g(x) u_t) \right] \\ J(x_T, T) &= c_T(x_T) \end{aligned} \quad (16)$$

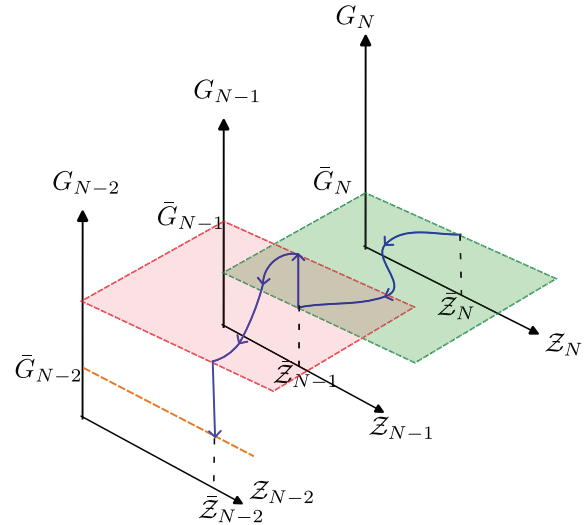


Fig. 2. Illustration of how the state and costate evolve backward in time. Since the actual dynamical system is continuous, the state changes continuously between time steps. The costate, however, remains constant between time steps and changes only at discrete points in time.

where  $J(x, t)$  is the optimal cost-to-go function from state  $x$  at time  $t$ . The optimal control is given by  $u_t = -(g(x)/r)(\partial J / \partial x)$ . Substituting the optimal control in (16) gives the PDE

$$\frac{\partial J}{\partial t} + l(x) - \frac{1}{2} \frac{g(x)^2}{r} \left( \frac{\partial J}{\partial x} \right)^2 + f(x) \frac{\partial J}{\partial x} = 0$$

whose characteristic equations are

$$\begin{aligned} \dot{x} &= f(x) - \frac{g(x)^2}{r} q \\ \dot{q} &= -l^x + \frac{g(x)g^x}{r} q^2 - f^x q, \text{ where } q = \frac{\partial J}{\partial x}. \end{aligned}$$

*Remark 4:* It might seem that (13) and (15) are nothing but the necessary conditions for optimal control, which, indeed they are. However, they give a new viewpoint from DP to solve terminal value problems backward in time and connect them to classical optimal control, which typically solves them as two-point boundary value problems. In particular, the backward flow perspective introduces the dynamical system viewpoint and helps us establish the uniqueness of the backward flow given any terminal state/costate pair. Consequently, satisfying the necessary conditions is also sufficient, as we shall now proceed to show under relatively mild assumptions.

For the development in the remainder of this section, the following assumption is made.

*Assumption 4:* Let  $\{\bar{\mathcal{Z}}'$  and  $\bar{G}'\}$  be the current state and costate, respectively, and  $\bar{\mathcal{Z}}$  and  $\bar{G}$  be the state and costate in the previous time step, respectively. Given  $\bar{\mathcal{Z}}'$  and  $\bar{G}'$ , (13) has a unique solution for  $\bar{\mathcal{Z}}$ :  $\bar{\mathcal{Z}} = \mathcal{F}(\bar{\mathcal{Z}}', \bar{G}')$ , where  $\mathcal{F}(\cdot, \cdot)$  is an implicit function. Moreover, two different state–costate pairs— $\{\bar{\mathcal{Z}}'_{(1)}, \bar{G}'_{(1)}\}$  and  $\{\bar{\mathcal{Z}}'_{(2)}, \bar{G}'_{(2)}\}$ —cannot lead to the same  $\{\bar{\mathcal{Z}}, \bar{G}\}$  under (13) and (15).

See Fig. 3 for an illustration of the meaning of the above assumption.

In the continuous time context, this assumption means that the characteristic ODEs have a unique backward and, hence,

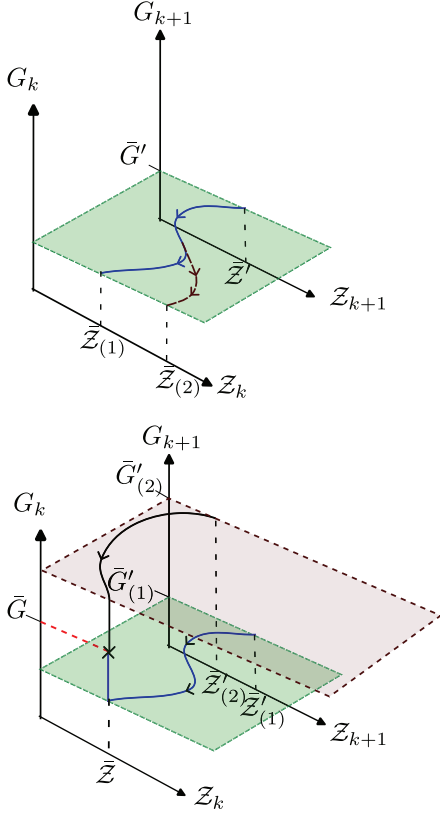


Fig. 3. Illustration of Assumption 4. In essence, the joint state/costate trajectory should not bifurcate or coalesce in the backward or, equivalently, forward path.

forward flow. A couple of important practical scenarios where the above assumption is satisfied are shown as follows.

**Lemma 3:** Let the functions,  $\mathcal{F}$  and  $\mathcal{G}$ ,  $l$  be twice continuously differentiable ( $\mathcal{C}^2$ ). If the underlying discretization time for the systems in (13) and (15) is sufficiently small and

$$\det[\bar{\mathcal{F}}^Z - (\mathcal{G}(\bar{Z})R^{-1}\nabla_Z\mathcal{G}(\bar{Z})^\top + \nabla_Z\mathcal{G}(\bar{Z})R^{-1}\mathcal{G}(\bar{Z})^\top)^\top \bar{G}'] \neq 0 \quad (17)$$

there is a unique map from  $\{\bar{Z}, \bar{G}\}$  to  $\{\bar{Z}', \bar{G}'\}$  and vice versa.

*Proof:* See Appendix C.  $\square$

The condition in (17) is interesting to note, and the fact that both the state equation in the backward map and the costate equation in the forward map gave the same condition is fascinating. It is the determinant of the closed-loop linear system governing the temporal perturbations (as opposed to the classical perturbations, which are spatial) around a linearizing point  $\{\bar{Z}, \bar{G}\}$ :  $\delta Z' = (A + BK)\delta Z$ , where

$$A = \mathcal{F}^Z|_{\bar{Z}} - \nabla_Z\mathcal{G}(\bar{Z})R^{-1}\mathcal{G}(\bar{Z})^\top \bar{G}, \quad B = \mathcal{G}(\bar{Z}) \\ K = -R^{-1}\nabla_Z\mathcal{G}(\bar{Z})^\top \bar{G}, \text{ and } \delta Z = \bar{Z} - \bar{Z}'.$$

**Lemma 4:** If the control influence matrix  $\mathcal{G}(\cdot)$  is state independent, then there is a unique map from  $\{\bar{Z}', \bar{G}'\}$  to  $\{\bar{Z}, \bar{G}\}$  and vice versa.

*Proof:* See Appendix D.  $\square$

Now, we show (13) and (15) that have a unique backward evolution and also show that trajectories originating from different terminal conditions do not cross.

**Lemma 5:** Under Assumption 4, given terminal conditions  $\{\bar{Z}_N, \bar{G}_N = c_N^Z|_{\bar{Z}_N}\}$ , the state and costate will have a unique evolution backward in time.

*Proof:* See Appendix E.  $\square$

**Lemma 6:** Under Assumption 4, trajectories originating from different terminal conditions do not cross in the  $\{\mathcal{Z}, \mathcal{G}\}$  space for all time.

*Proof:* See Appendix F.  $\square$

Using the results developed above, we show an important result in the following theorem that establishes that the problem in (8) has a unique solution. Hence, the open-loop control obtained by satisfying the necessary conditions is the globally optimal solution.

**Theorem 3 (Global Optimality of Open-Loop Solution)** Let the cost functions  $l(\cdot)$  and  $c_N(\cdot)$ , the drift  $\mathcal{F}(\cdot)$ , and the input influence function  $\mathcal{G}(\cdot)$  be  $\mathcal{C}^2$ , i.e., twice continuously differentiable. Under Assumption 1–4, an optimal trajectory that satisfies the MP (2) from a given initial state  $Z_0$  is the unique global minimum of the open-loop problem in (8).

*Proof:* See Appendix G.  $\square$

## B. Optimal Perturbation Feedback

In this section, we are interested in computing the optimal perturbation feedback. Suppose that given the optimal solution  $\bar{u}_k$  and the trajectory  $\bar{Z}_k$ ,  $k \in \{0, 1, \dots, N\}$  for a given initial condition  $\bar{Z}_0$ , from solving the open-loop optimal control problem. Let the nominal terminal state be  $\bar{Z}_N$ . We now expand the cost-to-go around this nominal optimal solution. To this purpose, let  $Z_k = \bar{Z}_k + \delta Z_k$ , for  $k \in \{0, 1, \dots, N\}$ . Then, expanding the optimal cost function around the nominal yields:  $J_k(Z_k) = \bar{J}_k + \bar{G}_k^\top \delta Z_k + (1/2)\delta Z_k^\top \bar{P}_k \delta Z_k + \dots$ , where  $\bar{J}_k = J_k(\bar{Z}_k)$ ,  $\bar{G}_k = \nabla_Z J_k|_{\bar{Z}_k}$ , and  $\bar{P}_k = \nabla_{ZZ}^2 J_k|_{\bar{Z}_k}$ . We can also do a perturbation expansion of the DP equation (12) around the optimal trajectory and equate the LHS and RHS terms based on the powers of  $\delta Z$  and get the following equations (the detailed derivation is in Appendix H):

$$\bar{J}_k = l(\bar{Z}_k) + \frac{1}{2}\bar{u}_k^\top R\bar{u}_k + \bar{J}_{k+1} \quad (18)$$

$$\bar{G}_k = \bar{l}_k^Z + \mathcal{A}_k^\top \bar{G}_{k+1} \quad (19)$$

$$\bar{P}_k = \mathcal{A}_k^\top \bar{P}_{k+1} \mathcal{A}_k + \bar{l}_k^{ZZ} + \sum_{i=1}^n \left[ \bar{\mathcal{F}}_{k,i}^{ZZ} + \sum_{j=1}^{n_u} \bar{\Gamma}_{k,i}^{j,ZZ} \bar{u}_k^j \right] \bar{G}_{k+1}^\top - K_k^\top (R + B_k^\top \bar{P}_{k+1} B_k) K_k \quad (20)$$

where  $K_k$  is defined in (22) and the definition of the other terms is the same as shown in Theorem 2. We have shown only terms up to the second order of  $\delta Z_k$ , and higher order terms can be obtained similarly. The optimal linear perturbation feedback is summarized in the following result.

**Proposition 2:** Consider the problem in (8) and let the optimal open-loop solution be denoted as  $\bar{Z}_k$  and  $\bar{u}_k$ , which satisfies the necessary conditions in Theorem 2. Then, the optimal linear perturbation feedback is given by

$$u_k = \bar{u}_k + K_k \delta Z_k, \text{ where} \quad (21)$$

$$K_k = -(R + B_k^\top \bar{P}_{k+1} B_k)^{-1} \left[ \sum_{i=1}^n \bar{G}_{k,i}^{\mathcal{Z},\top} \bar{G}_{k+1}^i + B_k^\top \bar{P}_{k+1} A_k \right]. \quad (22)$$

Note that, this “optimal linear perturbation feedback” is different from the traditional LQR perturbation feedback [7]. The equation for  $\bar{P}$  has additional second-order terms from the dynamics, and the feedback gain  $K$  has the additional term  $-(R + B_k^\top \bar{P}_{k+1} B_k)^{-1} \sum_{i=1}^n \bar{G}_{k,i}^{\mathcal{Z},\top} \bar{G}_{k+1}^i$ , which are absent in the LQR perturbation design, making the LQR design suboptimal for nonlinear problems.

*Proof:* See Appendix H.  $\square$

## V. OPTIMAL PERTURBATION FEEDBACK SYNTHESIS USING PARTIALLY OBSERVED DATA-BASED ILQR (POD-ILQR)

We use a generalized ILQR algorithm [29], [35] to find the optimal perturbation feedback  $u_k = \bar{u}_k + K_k \delta \mathcal{Z}_k$  from above. The ILQR algorithm, in theory, owing to the MP, can be generalized to iteratively solve the nonlinear information-state optimal control problem in (8), but it requires a key system identification step to identify the local linear time-varying system based on input–output data. We use the algorithm proposed in [31] for linear time-varying system identification. Thus, in the development below, we first present the information state ILQR algorithm and next show how to identify the LTV system parameters required by ILQR to develop the data-driven generalization. We note that the POD-ILQR is a model-based RL approach as we use the rollout data to identify the local LTV model in order to find the improved perturbation feedback policy rather than directly finding the improved policy.

*Remark 5:* In recent work [26], it was shown that the ILQR algorithm converges to the global optimal solution under some mild assumptions on cost functions (quadratic in control) and system dynamics (affine in control). Thus, the ILQR generalization to the proposed information-state-based formulation would also result in convergence to the global optimal solution under the same assumptions.

### A. Information-State-Based Generalized ILQR Algorithm

The ILQR algorithm consists of a forward and a backward pass given the current estimate of the perturbation feedback control.

*Forward Pass:* Given a nominal control sequence  $\{u_k\}_{k=0}^{N-1}$  and initial information-state vector  $\mathcal{Z}_0$ , the state is propagated in time in accordance with the dynamics  $\mathcal{Z}_{k+1} = \mathcal{F}(\mathcal{Z}_k) + \mathcal{G}(\mathcal{Z}_k)u_k$ . Thus, we get the nominal information-state trajectory  $(\bar{\mathcal{Z}}_k, \bar{u}_k)$ .

*Local LTV System:* Next, we find the corresponding local LTV system around the trajectory  $(\bar{\mathcal{Z}}_k, \bar{u}_k)$ , which can be written as:  $\delta \mathcal{Z}_{k+1} = A_k \delta \mathcal{Z}_k + B_k \delta u_k$ , where  $A_k$  and  $B_k$  are the linearization of the information-state dynamics about the nominal trajectory and  $\delta \mathcal{Z}_k = (\delta z_k, \delta z_{k-1}, \dots, \delta z_{k-q}, \delta u_{k-1}, \dots, \delta u_{k-q})$ , where  $\delta z_k = z_k - \bar{z}_k$  and  $\delta u_k = u_k - \bar{u}_k$  are the deviation from the nominal observation at time  $k$ .

*Backward Pass:* Given the LTV system parameters  $(A_k, B_k)$  around the nominal system, the ILQR algorithm computes a local optimal control as:  $\delta u_k = -\kappa_k - K_k \delta \mathcal{Z}_k$ , where  $\kappa_k = (R + B_k^\top V_{k+1} B_k)^{-1} (R \bar{u}_k + B_k^\top v_{k+1})$  and  $K_k = (R + B_k^\top V_{k+1} B_k)^{-1} B_k^\top V_{k+1} A_k$ , and

$$v_k = I_k^{\mathcal{Z}} + A_k^\top v_{k+1} - A_k^\top V_{k+1} B_k (R + B_k^\top V_{k+1} B_k)^{-1} \cdot (B_k^\top v_{k+1} + R \bar{u}_k) \quad (23)$$

$$V_k = I_k^{\mathcal{Z}\mathcal{Z}} + A_k^\top V_{k+1} A_k - A_k^\top V_{k+1} B_k (R + B_k^\top V_{k+1} B_k)^{-1} \cdot B_k^\top V_{k+1} A_k \quad (24)$$

with the terminal conditions  $v_N = (\partial c_N / \partial \mathcal{Z})|_{\mathcal{Z}_N}$  and  $V_N = \nabla_{\mathcal{Z}\mathcal{Z}}^2 c_N|_{\mathcal{Z}_N}$ . To compute  $v_k$  and  $V_k$ , we need their corresponding values in the future, i.e.,  $v_{k+1}$  and  $V_{k+1}$ . Given the terminal conditions from the forward pass and the local LTV model parameters  $(A_k$  and  $B_k)$ , we can do a backward-in-time sweep to compute the sequence  $v_k$  and  $V_k$  and, thus, the corresponding optimal control for that trajectory. (Notice the difference between (23) and (24) and (19) and (20). The difference is due to ignoring the second-order terms and the fact that the current trajectory  $\bar{u}$  need not satisfy the necessary conditions.)

*Update Trajectory:* Given the gains from the backward pass, we can update the nominal control sequence as

$$\bar{u}_k^{(i+1)} = \bar{u}_k^{(i)} + \kappa_k + K_k (\mathcal{Z}_k^{(i+1)} - \mathcal{Z}_k^{(i)}) \quad (25)$$

$$\kappa_k = (R + B_k^\top V_{k+1} B_k)^{-1} (R \bar{u}_k + B_k^\top v_{k+1}) \quad (26)$$

$$K_k = (R + B_k^\top V_{k+1} B_k)^{-1} B_k^\top V_{k+1} A_k \quad (27)$$

where  $\mathcal{Z}_0^{(i+1)} = \mathcal{Z}_0^{(i)}$ . We reiterate the process until  $R \bar{u}_k + B_k^\top v_{k+1} \approx 0$ . Thus, at convergence, we have the perturbation feedback law:

$u_k(\mathcal{Z}_k) = \bar{u}_k + K_k \delta \mathcal{Z}_k$ , where  $\delta \mathcal{Z}_k = \mathcal{Z}_k - \bar{\mathcal{Z}}_k$ , in which  $\bar{\mathcal{Z}}_k$  is the nominal information state evaluation under the optimal control  $\bar{u}_k$ .

Albeit, in theory, the ILQR is easily generalized to the partially observed case, the key difficulty is to find the information-state LTV system parameters  $A_k, B_k$ . In general, we cannot evaluate these system parameters analytically, and thus, we need to evaluate them in a data-based fashion. Fortunately, this may be accomplished using the previously proposed linear time varying-ARMA (LTV-ARMA) identification algorithm [31], which we recount in the following.

### B. Partially Observed Data-Based ILQR (POD-ILQR)

Let the nominal state be denoted by  $\bar{x}_k$  and the deviations from the nominal state as  $\delta x_k$ , which can be modeled as the following LTV system linearized around the nominal trajectory as:  $\delta x_k = A_{k-1} \delta x_{k-1} + B_{k-1} \delta u_{k-1}$ ,  $\delta z_k = C_k \delta x_k$ . First, we have the following result regarding the validity of an LTV ARMA model (see [31] for proof).

*Proposition 3:* An ARMA model of the order  $q$  given by  $\delta z_k = \alpha_{k-1} \delta z_{k-1} + \dots + \alpha_{k-q} \delta z_{k-q} + \beta_{k-1} \delta u_{k-1} + \dots + \beta_{k-q} \delta u_{k-q}$  exactly fits the LTV system given above if the matrix

$$O^q = [A_{k-q}^\top \dots A_{k-2}^\top C_{k-1}^\top, \dots, A_{k-q}^\top C_{k-q+1}^\top, C_{k-q}^\top]^\top$$



is full column rank. The exact parameters that match the LTV system can then be written as

$$\begin{aligned} & \begin{bmatrix} \alpha_{k-1} & | & \alpha_{k-2} & | \cdots & | & \alpha_{k-q} \end{bmatrix} \\ &= C_k A_{k-1} \dots A_{k-q} O^{q+} \\ & \begin{bmatrix} \beta_{k-1} & | & \beta_{k-2} & | \cdots & | & \beta_{k-q} \end{bmatrix} \\ &= -C_k A_{k-1} \dots A_{k-q} O^{q+} G^q \\ &+ \begin{bmatrix} C_k B_{k-1} & C_k A_{k-1} B_{k-2} & \cdots & C_k A_{k-1} \dots B_{k-q} \end{bmatrix} \end{aligned}$$

where matrix

$$G^q = \begin{bmatrix} 0 & C_{k-1} B_{k-2} & \cdots & \cdots & C_{k-1} A_{k-2} \dots B_{k-q} \\ 0 & 0 & C_{k-2} B_{k-3} & \cdots & C_{k-2} A_{k-3} \dots B_{k-q} \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & C_{k-q+1} B_{k-q} \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix}.$$

Notice that there always exists an exact fit for the ARMA model for any  $q > \bar{q}$ , if there exists a number  $\bar{q}$  for which the matrix  $O^{\bar{q}}$  is full column rank. Using this result, we can write linearized models in terms of ARMA parameters at each step along the nominal trajectory.

1) *Data-Driven Implementation*: Next, we show how to find the LTV ARMA parameters in a data-driven fashion using rollouts of the system.

*Nominal Trajectory*. First, we simulate the system with zero noise from the initial state  $Z_0$  using the updated control law (25) to get the updated nominal observation sequence  $\bar{z}_k$ .

*LTV-ARMA System Identification*. We perturb the linear system about the nominal trajectory and use the standard least-squares method to estimate the system parameters  $\alpha_{k-i}$  and  $\beta_{k-i}$  for  $i = 1, \dots, q$  from

$$\begin{aligned} \delta z_k^{(j)} &= \alpha_{k-1} \delta z_{k-1}^{(j)} + \cdots + \alpha_{k-q} \delta z_{k-q}^{(j)} \\ &+ \beta_{k-1} \delta u_{k-1}^{(j)} + \cdots + \beta_{k-q} \delta u_{k-q}^{(j)} \end{aligned} \quad (28)$$

where  $\delta z_k^{(j)} = z_k^{(j)} - \bar{z}_k$  is the perturbation from the nominal observation  $\bar{z}_k$ , and  $\delta u_k^{(j)} \sim \mathcal{N}(0, \sigma I)$  is the control input perturbation at step  $k$  for the  $j$ th simulation. These experiments can be done either in an open loop or closed loop. If done in open loop, we only add the perturbations  $\delta u_k^{(j)}$  above. If done in closed loop, we also add the feedback correction term  $\delta u_k^{(f,j)} = K_k \delta Z_k^{(j)}$ , and the total perturbation is  $\delta u_k^{(tot,j)} = \delta u_k^{(j)} + \delta u_k^{(f,j)}$ . The final linear relation between input-output perturbation and system parameters can be written as  $\mathbb{Z} = [\alpha_{k-1} \mid \alpha_{k-2} \mid \cdots \mid \alpha_{k-q} \mid \beta_{k-1} \mid \beta_{k-2} \mid \cdots \mid \beta_{k-q}] \mathbb{X}$ , where  $\mathbb{Z} = [\delta z_k^{(1)} \delta z_k^{(2)} \cdots \delta z_k^{(R)}]$  and

$$\mathbb{X} = \begin{bmatrix} \delta z_{k-1}^{(1)} & \delta z_{k-1}^{(2)} & \cdots & \delta z_{k-1}^{(R)} \\ \delta z_{k-2}^{(1)} & \delta z_{k-2}^{(2)} & \cdots & \delta z_{k-2}^{(R)} \\ \vdots & \vdots & \ddots & \vdots \\ \delta w_{k-1}^{(1)} & \delta w_{k-1}^{(2)} & \cdots & \delta w_{k-1}^{(R)} \\ \vdots & \vdots & \ddots & \vdots \\ \delta w_{k-q}^{(1)} & \delta w_{k-q}^{(2)} & \cdots & \delta w_{k-q}^{(R)} \end{bmatrix}.$$

The control perturbations  $\delta w_k^{(j)} = \delta u_k^{(j)}$  if the rollouts are done in open loop and equal  $\delta u_k^{(tot,j)}$  if the experiments are done

in closed loop. This can be solved using the standard least-squares method as:  $[\alpha_{k-1} \mid \cdots \mid \alpha_{k-q} \mid \beta_{k-1} \mid \cdots \mid \beta_{k-q}] = \mathbb{Z} \mathbb{X}^\dagger$ , where  $\mathbb{X}^\dagger$  denotes the pseudoinverse of some matrix  $\mathbb{X}$ . The correct length of the information state  $q$  is found by using the data length where the data matrix  $\mathbb{X}$  loses row rank, which follows from [31].

After identifying the system parameters  $\alpha_{k-1}, \dots, \alpha_{k-q}$  and  $\beta_{k-1}, \dots, \beta_{k-q}$  for  $k = \{0, \dots, N\}$ , we write the perturbation LTV system in the information state as given in (29), as shown at the bottom of the next page, which is written with the observation model as  $\delta Z_k = A_{k-1} \delta Z_{k-1} + B_{k-1} \delta u_{k-1}$ ,  $\delta z_k = C_k \delta Z_k$ , with  $C_k = [I_{n_z} \ 0]$ , where  $n_z$  is the number of measured outputs. Then, we use the identified  $A_{k-1}$  and  $B_{k-1}$  to find the optimal nominal trajectory using ILQR steps given earlier with alternating between backward and forward passes until convergence. The algorithms for the steps in POD-ILQR are given as follows.

### C. Why Is PODILQR Model-Based RL?

In a basic sense, if we understand RL as the process via which a feedback law is extracted from the input-output data of a dynamical system, then PODILQR is indeed a purely data-driven procedure that extracts the perturbation feedback law:  $u_k(Z_k) = \bar{u}_k + K_k \delta Z_k$  from the input-output data of the system.

However, a closer look reveals that it has the familiar actor-critic structure of typical RL algorithms. The “forward and backward pass equations” correspond to the “policy evaluation/critic” step using the improved perturbation feedback policy, while the “update trajectory” step corresponds to the “policy improvement/actor” step of the policy iteration approach. Since we search over a linear perturbation feedback parametrization of the control policy, it naturally leads to the ILQR equations, i.e., the ILQR is the most efficient way to search over this parametrization. The key step to make it purely data-driven is to identify the system linearization around the current nominal trajectory:  $(A_k, B_k)$ , and we accomplish via the LTV ARMA approach outlined above. Since we are explicitly identifying the LTV system around the current trajectory to improve the policy, the approach is “model-based.” It is left to future work to see if a model-free version of the same can be found.

## VI. POD2C ALGORITHM

In the previous section, we showed how to find the optimal nonlinear trajectory using the POD-ILQR algorithm. It was assumed that the system model used for training was noise-free. When the algorithm is tested in practice, process and measurement noises are inherent in the system, and in Section VII, we show how we can deal with them. We call this procedure the partially observed decoupled data-based control (POD2C) algorithm. It should be noted that POD2C is a heuristic algorithm, and no theoretical guarantees are provided, unlike previous sections. Though it is a heuristic, the empirical results show that it handles noise well. We assume that both the process and measurement noise are Gaussian and that their statistics are known. We consider process noise that comes through the actuator.



The POD2C algorithm proposes a two-step procedure to approximate the solution for the control problem in the presence of process and sensor noise.

*Step 1:* Solve a noiseless open-loop optimization problem to find an optimal control sequence,  $\bar{u}_k$ , and the optimal linear feedback gain  $K_k$  via the POD-ILQR scheme.

*Step 2:* Design an LQG estimator for the linear time-varying system around the optimal nominal trajectory

$$\delta Z_k = \mathcal{A}_{k-1} \delta Z_{k-1} + \mathcal{B}_{k-1} (\delta u_{k-1} + w_{k-1}) \quad (30a)$$

$$\delta z_k = \mathcal{C}_k \delta Z_k + v_k \quad (30b)$$

where  $w_k \sim \mathcal{N}(0, Q)$  and  $v_k \sim \mathcal{N}(0, R)$ .

We already showed how we solve the nominal trajectory in the previous section. Since there is noise, one needs a local feedback controller around the trajectory. Fortunately, the POD-ILQR scheme also provides the linear feedback law  $K_k$  for the LTV system. Under the assumption that the perturbations are small and the LTV approximation holds in that neighborhood, and assuming that the process and measurement noise are Gaussian, owing to certainty equivalence for linear systems,  $K_k$  is optimal for the noisy LTV system in (30).

Designing the estimator is not so straightforward since the initial uncertainty for  $\delta Z_0$  is unknown. However, a good estimator converges even with an incorrect initial uncertainty. Since (30) is linear, we invoke the separation principle for linear systems and design the estimator independently. We use the Kalman filter here since the system is linear, and the noises are Gaussian. To initialize the filter, we take the initial covariance to be the block diagonal of measurement noise covariance, i.e.,  $P_0 = \text{diag}([R, R, \dots, R])$ . The steps are shown in Algorithm 4.

*Remark 6:* Our approach assumes that we have a noise-free black box model (e.g., through a simulator) to find the optimal policy using the POD-ILQR algorithm. This is typical in RL applications, where the algorithm is trained on a noise-free system. The local linear controller and estimator in POD2C help to apply the optimal policy in practical cases where there is noise and is successful, as shown in the results. The question of whether we could learn the optimal policy from a system with noise is not within the scope of this article and is addressed in another paper here [36].

## VII. EMPIRICAL RESULTS

We use a black box MuJoCo simulator [37] to design the nominal trajectory and closed-loop feedback policy for the system with initial configurations given in Fig. 1.

### A. Cart-Pole

The classic robotics model is given in [28].

### B. 15-Link Swimmer

The system has 17 DOF and together with their rates and controls that are torques on the 14 joints.

### C. Fish

The fish has 13 DOF and six control channels with torques on the fin and tail joints.

### D. T2D1 Robotic Arm

The tensegrity model consists of 33 bars and 46 strings with controls as tensions in the strings.

### E. 1-D Viscous Burgers PDE

$(\partial u / \partial t) + u(\partial u / \partial x) = \nu(\partial^2 u / \partial x^2)$ . We take external control inputs on the boundaries corresponding to blowing/suction such that  $u(0, t) = U_1(t)$ ;  $u(L, t) = U_2(t)$ . Given an initial velocity profile  $u(x, 0) = \sin(\pi(x - 1))$ , the control task is to reach the goal state  $u(x, T) = -0.50$  for  $x = [0 : 0.02 : 2]$  with two observers added at both ends of the boundary.

### F. Allen-Cahn PDE

$(\partial \phi / \partial t) = -M((\partial F / \partial \phi) - \gamma \nabla^2 \phi)$ . We adopt the following general form of energy density function  $F$ :  $F(\phi; T, h) = \phi^4 + T\phi^2 + h\phi$ , where  $\phi(x, t) \in [-1, 1]$  is the order parameter and  $T(x, t)$  and  $h(x, t)$  are external control inputs. A periodic boundary condition is applied, and the control scheme is such that all grid points converging to the same goal state  $\phi_k = 1$  or  $-1$  are fed the same control inputs. Given a uniform initial condition  $\phi_0 = 0$ , the control task is to reach the custom goal state, as shown in Fig. 4.

Table I provides the time comparison to generate an optimal open-loop trajectory between the gradient descent [28] and the

$$\underbrace{\begin{bmatrix} \delta z_k \\ \delta z_{k-1} \\ \delta z_{k-2} \\ \vdots \\ \delta z_{k-q+1} \\ \delta u_{k-1} \\ \delta u_{k-2} \\ \delta u_{k-3} \\ \vdots \\ \delta u_{k-q+1} \end{bmatrix}}_{\delta Z_k} = \underbrace{\begin{bmatrix} \alpha_{k-1} & \alpha_{k-2} & \cdots & \alpha_{k-q+1} & \alpha_{k-q} & \beta_{k-2} & \beta_{k-3} & \cdots & \beta_{k-q+1} & \beta_{k-q} \\ 1 & 0 & \cdots & 0 & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & & \ddots & & \vdots & \vdots & & \ddots & \vdots & 0 \\ 0 & 0 & \cdots & 1 & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & & \ddots & & \vdots & \vdots & & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 & 0 & \cdots & 1 & 0 \end{bmatrix}}_{\mathcal{A}_{k-1}} \underbrace{\begin{bmatrix} \delta z_{k-1} \\ \delta z_{k-2} \\ \vdots \\ \delta z_{k-q+1} \\ \delta z_{k-q} \\ \delta u_{k-2} \\ \delta u_{k-3} \\ \vdots \\ \delta u_{k-q+1} \\ \delta u_{k-q} \end{bmatrix}}_{\delta Z_{k-1}} + \underbrace{\begin{bmatrix} \beta_{k-1} \\ 0 \\ \vdots \\ 0 \\ 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}}_{B_{k-1}} \delta u_{k-1} \quad (29)$$

**Algorithm 1** POD-ILQR Algorithm

---

```

1: Initialization:
2: Set initial trajectory  $\mathbb{T}_0$ , line search parameter  $\alpha$ , iteration
   counter  $i = 0$ .
3: while not converged do
4:   Step 1 - Backward Pass
5:    $k = N - 1$ .
6:   while  $k \geq 0$  do  $\triangleright$  LTV system ID using ARMA model
      $\triangleright$ 
7:      $\mathcal{A}_k, \mathcal{B}_k = \text{ARMA\_SysID}(\delta \mathcal{Z}_k, \delta \mathcal{Z}_{k-1}, \bar{u}_{k-1})$ 
8:      $\kappa_k = (R + \mathcal{B}_k^\top V_{k+1} \mathcal{B}_k)^{-1} (R \bar{u}_k + \mathcal{B}_k^\top v_{k+1})$ 
9:      $K_k = (R + \mathcal{B}_k^\top V_{k+1} \mathcal{B}_k)^{-1} \mathcal{B}_k^\top V_{k+1} \mathcal{A}_k$ 
10:     $v_k$  and  $V_k$  from Eqs. (23) and (24)
11:     $k = k - 1$ 
12:   end while
13:   return  $\{k_{0:N-1}, K_{0:N-1}\}$ 
14:   while cost reduction not acceptable do
15:     Reduce  $\alpha$ 
16:     Step 2 - Forward Pass
17:      $k = 0$ 
18:     while  $k < N$  do
19:        $\bar{u}_k = \bar{u}_k^{prev} + \alpha \kappa_k + K_k(\bar{\mathcal{Z}}_k - \bar{\mathcal{Z}}_k^{prev})$ 
20:        $\bar{x}_{k+1} = f(\bar{x}_k, \bar{u}_k)$ ,  $\bar{\mathcal{Z}}_{k+1} = h(\bar{x}_{k+1})$ 
21:        $J_i = J_i + \text{incremental\_cost}(\bar{\mathcal{Z}}_k, \bar{u}_k)$ 
22:        $k = k + 1$ 
23:     end while
24:      $J_i = J_i + \text{terminal\_cost}(\bar{\mathcal{Z}}_N)$ 
25:      $\mathbb{T}_{i+1} = \{\bar{\mathcal{Z}}_{0:N}, \bar{u}_{0:N-1}\}$ 
26:     return  $\{\mathbb{T}_{i+1}, J_i\}$ 
27:      $i = i + 1$ 
28:   end while
29: end while
30:  $\mathbb{T}^* = \mathbb{T}_{i+1}$ 

```

---

**Algorithm 2** Forward Pass

---

```

1: Input: Nominal trajectory  $\{\bar{\mathcal{Z}}_k^{prev}, \bar{u}_k^{prev}\} \leftarrow \mathbb{T}_k$ , ILQR gains
    $\{k_{0:N-1}, K_{0:N-1}\}$ 
2: Start from  $t = 0$ ,  $cost = 0$ ,  $\bar{x}_0 = x_0$ 
3: while  $t < N$  do
4:    $\bar{\mathcal{Z}}_k = C_k \bar{x}_k$ 
5:    $\bar{u}_k = \bar{u}_k^{prev} + \alpha \kappa_k + K_k(\bar{\mathcal{Z}}_k - \bar{\mathcal{Z}}_k^{prev})$ 
6:    $\bar{x}_{k+1} = \text{simulate\_one\_step}(\bar{x}_k, \bar{u}_k)$ 
7:    $cost = cost + \text{incremental\_cost}(\bar{\mathcal{Z}}_k, \bar{u}_k)$ 
8:    $t = t + 1$ 
9: end while
10:  $cost = cost + \text{terminal\_cost}(\bar{\mathcal{Z}}_N)$ 
11:  $\mathbb{T}_{k+1} = \{\bar{\mathcal{Z}}_{0:N}, \bar{u}_{0:N-1}\}$ 
12: return  $\mathbb{T}_{k+1}, cost$ 

```

---

proposed POD-ILQR. The output number in Table I represents the minimum number of measurements needed to fit a good  $q$ th order ARMA model. We observed that as the dimension of the system increases, the relative number of measurements required decreases. The cart's position and pole angle are used as feedback for the cart-pole model. The 15-link swimmer requires only angular positions of every other joint and the

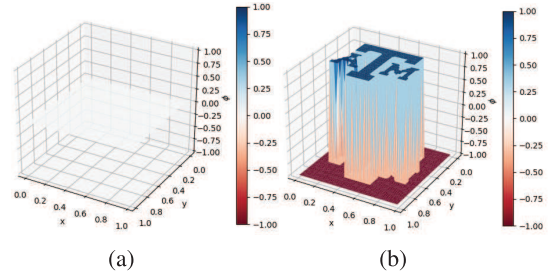


Fig. 4. High-dimensional, complex, and partially observed nature of the problem governed by Allen-Cahn PDE (2500 states and 16 outputs). (a) Initial state. (b) Final state.

TABLE I  
COMPARISON RESULTS BETWEEN GRADIENT DESCENT [28] AND POD-ILQR (\*GRADIENT DESCENT FAILED TO CONVERGE)

| System          | Grad. Des. Total Time (sec.) | POD-ILQR Total Time (sec.) | Iteration Number | $q$ | States | Outputs |
|-----------------|------------------------------|----------------------------|------------------|-----|--------|---------|
| Cart-pole       | 15                           | 0.8                        | 40               | 2   | 4      | 2       |
| Swimmer         | 376736.72                    | 2110.0                     | 100              | 5   | 34     | 10      |
| Fish            | *                            | 1720.5                     | 200              | 2   | 27     | 11      |
| Robotic Arm     | *                            | 355.8                      | 40               | 3   | 150    | 24      |
| Burgers Eqn.    | 2763.4                       | 38.2                       | 10               | 2   | 100    | 2       |
| Allen-Cahn Eqn. | 21078.18                     | 47.78                      | 10               | 2   | 2500   | 16      |

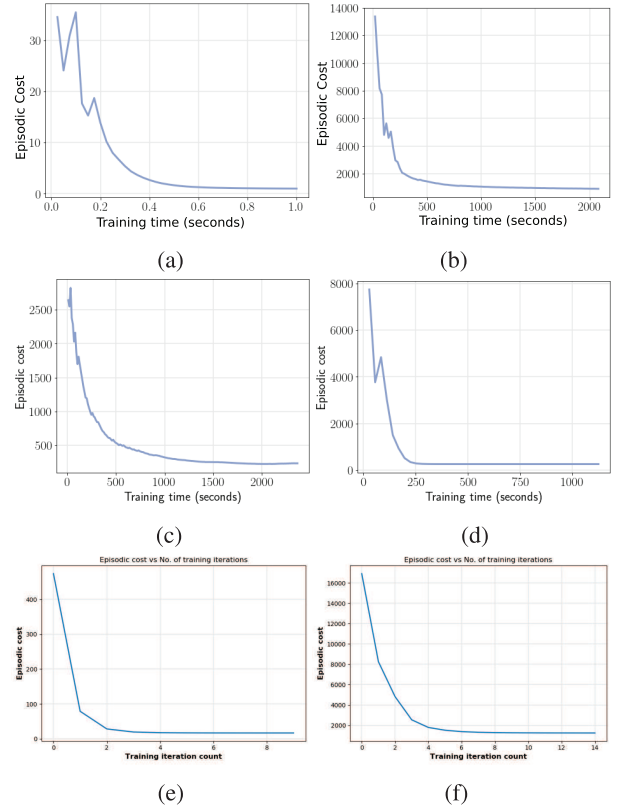


Fig. 5. Convergence of episodic cost during training. (a) Cart-pole. (b) 15-link swimmer. (c) Fish. (d) Robotic arm. (e) PDE-burgers. (f) PDE Allen-Cahn.

fish needs only the angles of the fin and tail joints. The  $T_2D_1$  arm is a soft-robotic arm where the node positions of 8 (out of 25) node points were used.

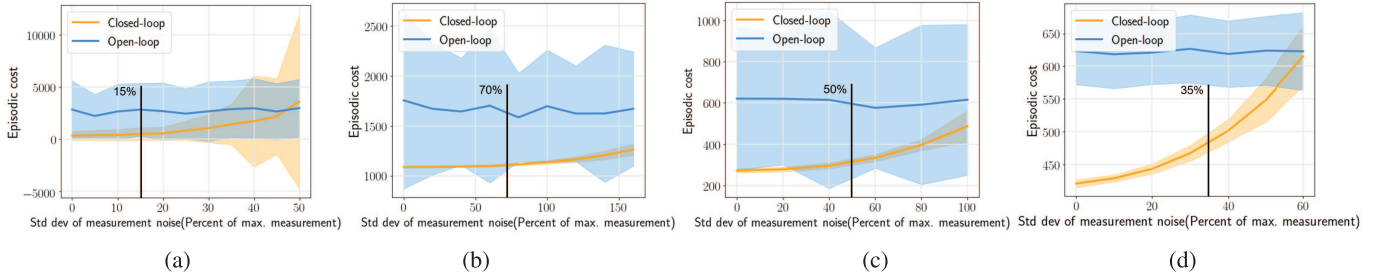


Fig. 6. Averaged episodic reward versus measurement noise level for fixed 10% process noise with LQG as closed-loop feedback (a) Cart-pole. (b) 15-link swimmer. (c) Fish. (d) T2D1 robotic arm.

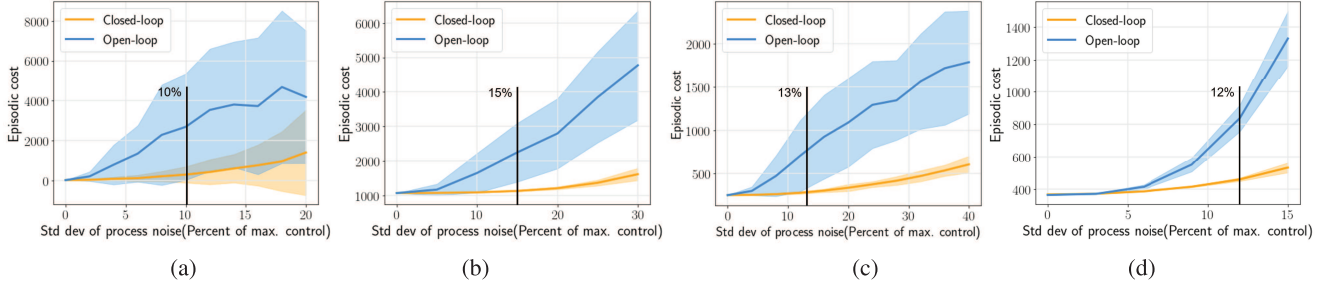


Fig. 7. Averaged episodic reward versus process noise level for fixed 10% measurement noise with LQG as closed-loop feedback (a) Cart-pole. (b) 15-link swimmer. (c) Fish. (d) T2D1 robotic arm.

### Algorithm 3 Backward Pass

```

1:  $t = N - 1$  ▷/* start from terminal step */
2: Compute  $J_{\mathcal{Z}_N}$  and  $J_{\mathcal{Z}_N \mathcal{Z}_N}$  using boundary conditions
3: while  $t \geq 0$  do ▷/* obtain the augmented Jacobians using ARMA model */
4:  $\mathcal{A}_k, \mathcal{B}_k \leftarrow \text{ARMA\_fitting}(\delta \mathcal{Z}_k, \delta \mathcal{Z}_{k-1}, u_{k-1})$  ▷/* obtain the partials of the Q function */
5:  $Q_{\mathcal{Z}_k} = c_{\mathcal{Z}_k} + \mathcal{A}_k^\top J'_{\mathcal{Z}_{k+1}}$ 
6:  $Q_{\mathbf{u}_k} = c_{\mathbf{u}_k} + \mathcal{B}_k^\top J'_{\mathcal{Z}_{k+1}}$ 
7:  $Q_{\mathcal{Z}_k \mathcal{Z}_k} = c_{\mathcal{Z}_k \mathcal{Z}_k} + \mathcal{A}_k^\top J'_{\mathcal{Z}_{k+1} \mathcal{Z}_{k+1}} \mathcal{A}_k$ 
8:  $Q_{\mathbf{u}_k \mathcal{Z}_k} = c_{\mathbf{u}_k \mathcal{Z}_k} + \mathcal{B}_k^\top (J'_{\mathcal{Z}_{k+1} \mathcal{Z}_{k+1}} + \mu I_{n_x \times n_x}) \mathcal{A}_k$ 
9:  $Q_{\mathbf{u}_k \mathbf{u}_k} = c_{\mathbf{u}_k \mathbf{u}_k} + \mathcal{B}_k^\top (J'_{\mathcal{Z}_{k+1} \mathcal{Z}_{k+1}} + \mu I_{n_x \times n_x}) \mathcal{B}_k$ 
10: if  $Q_{\mathbf{u}_k \mathbf{u}_k}$  is positive-definite then
11:    $\kappa_k = -Q_{\mathbf{u}_k \mathbf{u}_k}^{-1} Q_{\mathbf{u}_k}$ 
12:    $K_k = -Q_{\mathbf{u}_k \mathbf{u}_k}^{-1} Q_{\mathbf{u}_k \mathcal{Z}_k}$ 
13:   Decrease  $\mu$ 
14: else
15:   Increase  $\mu$ 
16:   Redo backward pass for current timestep
17: end if ▷/* obtain the partials of the value function  $J_k$  */
18:  $J_{\mathcal{Z}_k} = Q_{\mathcal{Z}_k} + K_k^\top Q_{\mathbf{u}_k \mathbf{u}_k} \kappa_k + K_k^\top Q_{\mathbf{u}_k} + Q_{\mathbf{u}_k \mathcal{Z}_k}^\top \kappa_k$ 
19:  $J_{\mathcal{Z}_k \mathcal{Z}_k} = Q_{\mathcal{Z}_k \mathcal{Z}_k} + K_k^\top Q_{\mathbf{u}_k \mathbf{u}_k} K_k + K_k^\top Q_{\mathbf{u}_k \mathcal{Z}_k} + Q_{\mathbf{u}_k \mathcal{Z}_k}^\top K_k$ 
20:  $t = t - 1$ 
21: end while
22: return  $\{\kappa_{0:N-1}, K_{0:N-1}\}$ 

```

### G. Nominal Trajectory Design With POD-ILQR

The 15-link swimmer and the fish take more iterations to converge as they have higher nonlinearity due to the fluid–structure interaction in the swimming motion. However, the training is far more time-efficient compared with the

### Algorithm 4 POD2C Algorithm

```

1: Initialization:  $P_0 = \text{diag}([R, R, \dots, R])$ 
2:  $\bar{u}_k, K_k \leftarrow \text{POD-ILQR algorithm}$ 
3:  $L_k \leftarrow \text{Kalman Filter}(\mathcal{A}_k, \mathcal{B}_k, C_k, Q, R, P_0)$ 
4: Estimated state:  $\delta \hat{\mathcal{Z}}_k = \mathcal{A}_{k-1} \delta \hat{\mathcal{Z}}_{k-1} + \mathcal{B}_{k-1} \delta u_{k-1} + L_k (\delta z_k - C_k \mathcal{A}_{k-1} \delta \hat{\mathcal{Z}}_{k-1} - C_k \mathcal{B}_{k-1} \delta u_{k-1})$ 
5: Control:  $\delta u_k = K_k \delta \hat{\mathcal{Z}}_k$ , and  $u_k = \bar{u}_k + \delta u_k$ 

```

first-order gradient descent method used in our prior work [28]. The T2D1 arm system also takes less time and iterations to converge despite the high dimensionality of the model and limited outputs. The training convergence is shown in Fig. 5. The iteration numbers given in Table I show smooth and efficient convergence, even for systems with high nonlinearity and high dimensionality. The results are obtained using MATLAB and MuJoCo on a Ryzen 3700 PC. The most time-consuming procedure is running simulations to collect data for fitting the ARMA model, which is in serial for now, but can be in parallel as the rollouts are independent of each other, further improve the time efficiency, and have the potential for real-time operation.

### H. Robustness to Measurement and Process Noise

Fig. 6 shows the plots for the episodic cost with the variation in the measurement noise with a fixed value of process noise. Similarly, Fig. 7 shows the plots for the episodic cost with the variation in the process noise for a fixed value of measurement noise. Fig. 6 compares the open-loop and the closed-loop control policies under different measurement noise levels, while the process noise standard deviation is set to 10% of the maximal nominal control. The measurement noise level on

the  $x$ -axis is the percentage of the measurement noise standard deviation with respect to the maximal nominal measurement. Note that both the measurement and process noise are added as zero-mean Gaussian i.i.d. noise to all measurement and control channels at each step.

Note that the closed-loop policy can successfully finish the task with smaller noise levels than what is indicated by the black threshold lines in the figure. Under the open-loop policy, the process noise drives the model off the nominal trajectory and results in high episodic cost, while the closed-loop feedback can help the system stay close to the nominal trajectory. This can be seen from the figure as the episodic cost mean and variance of the closed-loop policy is much smaller than the open-loop policy on the entire tested noise range, although both policies fail the task when the process noise becomes larger than what the black threshold line indicates. The threshold values of process and measurement noise are empirically calculated at which the system will go too far away from the nominal trajectory, and the final goal cannot be achieved. The above analysis regarding the performance of control policy under noise proves that the LQG closed-loop feedback wrapped around the nominal trajectory makes the full closed-loop policy robust to measurement and process noise.

### I. Comparison With a Direct RL Method

In a direct RL method such as DDPG [15], deep neural networks are used to represent the closed-loop control policy. In general, Direct RL methods require full-state observation, and it is not clear how to generalize them to partially observed problems. Nonetheless, we used the DDPG method on the cart-pole and fish examples with the same information state as found by POD2C. Though it worked for the cart-pole system, it failed to converge on high-dimensional systems like the fish. Even after training for approximately 20 h, the fish could not swim to the target.

## VIII. CONCLUSION

This article proposed an optimal information-state approach to transform partially observed problems into fully observed optimal nonlinear control problems. We showed that the information-state-based formulation is equivalent to the original system formulation with the partial nonlinear observation model. This article further provides the conditions to meet the MP in the information-state formulation. We then provide the algorithm that generated  $q$ th-order ARMA models using the input–output data to model LTV systems in the information state, which further allowed designing the optimal nominal trajectory using ILQR with partial-state observations. Finally, this article presented a decoupled data-based approach to control complex robotic systems by designing the closed-loop feedback law with partial-state observations. Empirical results are shown for complex robotic systems, including challenging cases of hard-to-model soft contact constraints, dynamic fluid–structure interactions, and PDEs, under motion as well as sensing uncertainty. In our opinion, the POD2C approach is highly efficient for RL in partially observed

problems; however, questions regarding optimality under noise will be explored in future work.

## APPENDIX

### A. Proof of Theorem 1

*Proof:* Consider the two problems in (5) and (6). To show the equivalence between the two problems, one needs to show: 1) the initial conditions are equivalent; 2) the input–output response of the state-space system and information-state system are equal; and 3) the optimal feedback policies of both systems generate the same control inputs.

Using Lemma 1, the initial condition for the systems are related by  $x_{\bar{q}} = \Psi(\mathcal{Z}_{\bar{q}}^q)$ , where the information state  $\mathcal{Z}_{\bar{q}}^q = [z_{\bar{q}}^T, \dots, z_0^T, u_{\bar{q}-1}^T, \dots, u_0^T]^T$ . Furthermore, Lemma 2 shows that they have the same input–output response.

To show that the feedback policies generate the same control inputs, let the optimal control policy mapping the state to the control at time  $k$  for the state-space and information-space system be  $\pi_x^k(\cdot)$  and  $\pi_{\mathcal{Z}}^k(\cdot)$ , respectively. The policy for the true full-state feedback  $\pi_x^k(\cdot)$  can be applied to the information-state system, by using the transformation from Lemma 1,  $x_k = \Psi(\mathcal{Z}_k)$ , which gives the control input to be  $u_k = \pi_x^k(\Psi(\mathcal{Z}_k))$ . Then, because of optimality of the policy  $\pi_{\mathcal{Z}}(\cdot)$  for the information-state system, the input–output equivalence of the systems in (5b) and (6b), and the same cost functions, we get  $J_{\mathcal{Z}}^{\pi_{\mathcal{Z}}} \leq J_x^{\pi_x}$ . Similarly, the optimal information-state policy  $\pi_{\mathcal{Z}}(\cdot)$  can be applied to the true state-space model by taking the past inputs and outputs as the argument to the policy. Because of optimality of  $\pi_x$  for the state-space system, we get  $J_x^{\pi_x} \leq J_{\mathcal{Z}}^{\pi_{\mathcal{Z}}}$ . Owing to the above two inequalities,  $J_x^{\pi_x} = J_{\mathcal{Z}}^{\pi_{\mathcal{Z}}}$ . Hence, the optimal feedback on the information state is identical to the optimal control for the underlying state-space system.  $\square$

### B. Proof of Proposition 1

*Proof:* We write the system dynamics with a forward Euler approximation for a small discretization time:  $x_{k+1} = x_k + f(x_k)\Delta t + g(x_k)u_k\Delta t$ , and the observation model as:  $z_k = h(x_k) = h(x_{k-1} + dx_k)$ , where  $dx_k = f(x_{k-1})\Delta t + g(x_{k-1})u_{k-1}\Delta t$ . Given that the time discretization is sufficiently small, the observation model can be written using a Taylor expansion as

$$\begin{aligned} z_k &= h(x_{k-1} + dx_k) = h(x_{k-1}) + \underbrace{\frac{\partial h}{\partial x}}_{H(x_{k-1})} dx_k + O(\Delta t^2) \\ z_k &= \underbrace{h(x_{k-1}) + H(x_{k-1})f(x_{k-1})\Delta t}_{F(x_{k-1})} \\ &\quad + \underbrace{H(x_{k-1})g(x_{k-1})\Delta t}_{G(x_{k-1})} u_{k-1} + O(\Delta t^2). \end{aligned}$$

Thus, we can write the observation  $z_k$  as some function of  $x_{k-1}$  and  $u_{k-1}$  as:  $z_k = F(x_{k-1}) + G(x_{k-1})u_{k-1} + O(\Delta t^2)$ .

Further substituting for  $x_{k-1}$  as  $x_{k-1} = \Psi(\mathcal{Z}_{k-1}^q)$  from above, we can write the  $z_k$  in terms of information state as

$$z_k = F(\Psi(\mathcal{Z}_{k-1}^q)) + G(\Psi(\mathcal{Z}_{k-1}^q))u_{k-1} + O(\Delta t^2). \quad (31)$$



For sufficiently small  $\Delta t$ , where  $O(\Delta t^2) \rightarrow 0$ , (31) can be written in terms of  $\mathcal{Z}_k^q$  in a control-affine form as shown in (7).  $\square$

### C. Proof of Lemma 3

*Proof:* The idea behind the proof is as follows. Since the discretization time  $\Delta t$  is sufficiently small, and owing to the smoothness of the underlying functions, the implicit function theorem [32] shows that, in a sufficiently small neighborhood of  $\{\tilde{\mathcal{Z}}', \tilde{G}'\}$  facilitated by the small  $\Delta t$ , there exists a unique  $\{\tilde{\mathcal{Z}}, \tilde{G}\}$ , that satisfies (13) and (15), if condition in (17) holds.

First, we will consider the solution to the equations (13) and (15) backward in time. If  $\Delta t$  is small,  $\{\tilde{\mathcal{Z}}, \tilde{G}\}$  will be in the neighborhood of  $\{\tilde{\mathcal{Z}}', \tilde{G}'\}$ , i.e.,  $\tilde{\mathcal{Z}} = \tilde{\mathcal{Z}}' + \delta\mathcal{Z}$ , where  $\delta\mathcal{Z} = \delta\mathcal{Z}(\Delta t)$ . Using this relationship, (13) can be written as

$$\tilde{\mathcal{Z}}' = \mathcal{F}(\tilde{\mathcal{Z}}' + \delta\mathcal{Z}) - \mathcal{G}(\tilde{\mathcal{Z}}' + \delta\mathcal{Z}) R^{-1} \mathcal{G}(\tilde{\mathcal{Z}}' + \delta\mathcal{Z})^\top \tilde{G}'.$$

Doing a perturbation expansion around  $\tilde{\mathcal{Z}}'$  gives

$$\begin{aligned} & [\mathcal{F}^{\mathcal{Z}}|_{\tilde{\mathcal{Z}}'} - (\mathcal{G}(\tilde{\mathcal{Z}}') R^{-1} \nabla_{\mathcal{Z}} \mathcal{G}|_{\tilde{\mathcal{Z}}'})^\top \\ & \quad + \nabla_{\mathcal{Z}} \mathcal{G}|_{\tilde{\mathcal{Z}}'} R^{-1} \mathcal{G}(\tilde{\mathcal{Z}}')^\top]^\top \tilde{G}' \delta\mathcal{Z} \\ & = \tilde{\mathcal{Z}}' - \mathcal{F}(\tilde{\mathcal{Z}}') + \mathcal{G}(\tilde{\mathcal{Z}}') R^{-1} \mathcal{G}(\tilde{\mathcal{Z}}')^\top \tilde{G}' + O(\Delta t^2). \end{aligned} \quad (32)$$

For a sufficiently small  $\Delta t$ ,  $O(\Delta t^2) \rightarrow 0$ . Then, for a unique  $\delta\mathcal{Z}$  to exist, the following condition should be satisfied:

$$\det[\mathcal{F}^{\mathcal{Z}}|_{\tilde{\mathcal{Z}}'} - (\mathcal{G}(\tilde{\mathcal{Z}}') R^{-1} \nabla_{\mathcal{Z}} \mathcal{G}|_{\tilde{\mathcal{Z}}'})^\top + \nabla_{\mathcal{Z}} \mathcal{G}|_{\tilde{\mathcal{Z}}'} R^{-1} \mathcal{G}(\tilde{\mathcal{Z}}')^\top]^\top \tilde{G}' \neq 0. \quad (33)$$

If the above condition is satisfied,  $\delta\mathcal{Z}$  can be calculated by (32), and  $\tilde{\mathcal{Z}} = \tilde{\mathcal{Z}}' + \delta\mathcal{Z}$  is unique. Computation of  $\tilde{G}$  is straightforward since  $\tilde{G}$  is an explicit function of  $\tilde{\mathcal{Z}}$  and  $\tilde{G}'$  as given in (15). Hence,  $\tilde{\mathcal{Z}}$  and  $\tilde{G}$  are unique in the backward direction.

Similarly, in the forward direction, given  $\{\tilde{\mathcal{Z}}, \tilde{G}\}$ , we need to first compute  $\tilde{G}'$  from (15). Using the similar arguments as done above, for sufficiently small  $\Delta t$ ,  $\tilde{G}' = \tilde{G} + \delta G$ , where  $\delta G = \delta G(\Delta t)$ . Then, (15) can be written as

$$\begin{aligned} \tilde{G} &= \tilde{L}^{\mathcal{Z}} + (\mathcal{F}^{\mathcal{Z}}|_{\tilde{\mathcal{Z}}} \\ & \quad - \nabla_{\mathcal{Z}} \mathcal{G}|_{\tilde{\mathcal{Z}}} R^{-1} \mathcal{G}(\tilde{\mathcal{Z}})^\top (\tilde{G} + \delta G))^\top (\tilde{G} + \delta G). \end{aligned}$$

Doing a perturbation expansion around  $\tilde{G}$  gives

$$\begin{aligned} & [\mathcal{F}^{\mathcal{Z}}|_{\tilde{\mathcal{Z}}} - (\mathcal{G}(\tilde{\mathcal{Z}}) R^{-1} \nabla_{\mathcal{Z}} \mathcal{G}|_{\tilde{\mathcal{Z}}})^\top + \nabla_{\mathcal{Z}} \mathcal{G}|_{\tilde{\mathcal{Z}}} R^{-1} \mathcal{G}(\tilde{\mathcal{Z}})^\top]^\top \tilde{G} \\ & = \tilde{G} - \tilde{L}^{\mathcal{Z}} - (\mathcal{F}^{\mathcal{Z}}|_{\tilde{\mathcal{Z}}} - \nabla_{\mathcal{Z}} \mathcal{G}|_{\tilde{\mathcal{Z}}} R^{-1} \mathcal{G}(\tilde{\mathcal{Z}})^\top)^\top \tilde{G} + O(\Delta t^2). \end{aligned}$$

For a sufficiently small  $\Delta t$ ,  $O(\Delta t^2) \rightarrow 0$ . Then, for a unique  $\delta G$ , the following condition should be satisfied:

$$\det[\mathcal{F}^{\mathcal{Z}}|_{\tilde{\mathcal{Z}}} - (\mathcal{G}(\tilde{\mathcal{Z}}) R^{-1} \nabla_{\mathcal{Z}} \mathcal{G}|_{\tilde{\mathcal{Z}}})^\top + \nabla_{\mathcal{Z}} \mathcal{G}|_{\tilde{\mathcal{Z}}} R^{-1} \mathcal{G}(\tilde{\mathcal{Z}})^\top]^\top \tilde{G} \neq 0. \quad (34)$$

Then,  $\tilde{G}'$  is given by  $\tilde{G}' = \tilde{G} + \delta G$  and  $\tilde{\mathcal{Z}}'$  can be computed from (13) and is unique since it is an explicit function. Notice that the condition for a unique map for the forward as well as the backward evolution is identical but evaluated at different time steps. Hence, (13) and (15) have unique solutions.  $\square$

### D. Proof of Lemma 4

*Proof:* Given  $\{\tilde{\mathcal{Z}}', \tilde{G}'\}$ ,  $\bar{u}$  is exactly determined from (9) without the knowledge of  $\tilde{\mathcal{Z}}$  since  $\mathcal{G}(\cdot)$  is state independent. Since the state reached by a dynamical system  $\tilde{\mathcal{Z}}'$ , and the control action  $\bar{u}$  taken to reach there are known, the state it originated from ( $\tilde{\mathcal{Z}}$ ) is unique. Also, (15) is a linear equation for state independent  $\mathcal{G}$ ; hence,  $\tilde{G}$  is unique. The uniqueness of the forward map can be similarly shown.  $\square$

### E. Proof of Lemma 5

*Proof:* Let us consider at  $k = N - 1$ . Since  $\{\tilde{\mathcal{Z}}_N, \tilde{G}_N\}$  are given,  $\tilde{\mathcal{Z}}_{N-1}$  is unique due to Assumption 4. Then, it is straightforward to see that  $\tilde{G}_{N-1}$  is unique from (15). The same argument is applicable at every time step  $k$ .  $\square$

### F. Proof of Lemma 6

*Proof:* The only scenario the trajectories might cross is in between the time steps. The state of the dynamical system evolves continuously in time between time steps, while the costate remains constant (see Fig. 2). If trajectories have to intersect, they should start with the same costate, but different states. At the crossing point, they will have the same state, and a dynamical system cannot have different evolutions from the same state and costate.  $\square$

### G. Proof of Theorem 3

*Proof:* We show the scalar case; the vector case is a straightforward extension. From Lemma 5, the state and costate equations (13) and (15) have unique backward solutions for  $k \in \{0, 1, \dots, N\}$  from the terminal value  $(\mathcal{Z}_N, G_N = c_T^{\mathcal{Z}}(\mathcal{Z}_N))$ . Note that the terminal costate  $G_N$  is a function of  $\mathcal{Z}_N$ , i.e.,  $G_N$  is uniquely determined by the value of  $\mathcal{Z}_N$ .

The family of curves originating from the terminal points  $(\mathcal{Z}_N, G_N)$  on the boundary forms the general solution to the DP problem (12), and they do not cross as shown in Lemma 6. Hence, the trajectory that passes through the initial state  $\mathcal{Z}_0$  satisfies Pontryagin's MP (PMP) for the discrete-time optimal control problem, and the corresponding  $G_0 = (\partial J / \partial \mathcal{Z})|_{\mathcal{Z}_0}$  is the initial costate. Under Assumption 3, the optimal cost-to-go function  $J_k(\cdot)$  is differentiable. Hence, the costate  $G_0 = (\partial J / \partial \mathcal{Z})|_{\mathcal{Z}_0}$  corresponding to the initial state  $\mathcal{Z}_0$  is unique, as otherwise the solution would not be differentiable at  $\mathcal{Z}_0$ , and thus, the characteristic curve passing through it is the global optimum solution. This establishes the fact that if the DP problem admits a smooth solution, the PMP is also sufficient. Moreover, this also shows that if the DP problem has a  $C^1$  solution, it is also unique since it is completely characterized by the (unique) curves flowing from the terminal state/costate pair  $(\mathcal{Z}_N, G_N)$ .  $\square$

### H. Proof of Proposition 2

*Proof:* Let  $\mathcal{Z}_k = \tilde{\mathcal{Z}}_k + \delta\mathcal{Z}_k$ , for  $k \in \{0, 1, \dots, N\}$ . Then, expanding the optimal cost function around the nominal yields:  $J_k(\mathcal{Z}_k) = \bar{J}_k + \tilde{G}_k^\top \delta\mathcal{Z}_k + (1/2) \delta\mathcal{Z}_k^\top \bar{P}_k \delta\mathcal{Z}_k + \dots$ , where  $\bar{J}_k = J_k(\tilde{\mathcal{Z}}_k)$ ,  $\tilde{G}_k = \nabla_{\mathcal{Z}} J_k|_{\tilde{\mathcal{Z}}_k}$ , and  $\bar{P}_k = \nabla_{\mathcal{Z}\mathcal{Z}} J_k|_{\tilde{\mathcal{Z}}_k}$ . Now, do

a perturbation expansion of the DP equation (12) around the optimal trajectory. Then, (12) can be written as

$$\begin{aligned} \bar{J}_k + \bar{G}_k^\top \delta \mathcal{Z}_k + \frac{1}{2} \delta \mathcal{Z}_k^\top \bar{P}_k \delta \mathcal{Z}_k + O(\delta \mathcal{Z}_k^3) \\ = \min_{\delta u_k} \left[ l(\bar{\mathcal{Z}}_k) + \bar{l}_k^{\mathcal{Z},\top} \delta \mathcal{Z}_k \right. \\ \left. + \frac{1}{2} \delta \mathcal{Z}_k^\top \bar{l}_k^{\mathcal{Z}\mathcal{Z}} \delta \mathcal{Z}_k + \frac{1}{2} \bar{u}_k^\top R \bar{u}_k + \bar{u}_k^\top R \delta u_k + \frac{1}{2} \delta u_k^\top R \delta u_k \right. \\ \left. + \bar{J}_{k+1} + \bar{G}_{k+1}^\top \delta \mathcal{Z}_{k+1} + \frac{1}{2} \delta \mathcal{Z}_{k+1}^\top \bar{P}_{k+1} \delta \mathcal{Z}_{k+1} + O(\delta \mathcal{Z}_k^3) \right]. \end{aligned} \quad (35)$$

Doing a perturbation expansion of the dynamics around  $\bar{\mathcal{Z}}_k$  and  $\bar{u}_k$  gives

$$\begin{aligned} \delta \mathcal{Z}_{k+1} = \mathcal{A}_k \delta \mathcal{Z}_k + \mathcal{B}_k \delta u_k + \frac{1}{2} \delta \mathcal{Z}_k^\top \sum_{i=1}^n \\ \times \left[ \bar{\mathcal{F}}_{k,i}^{\mathcal{Z}\mathcal{Z}} + \sum_{j=1}^{n_u} \bar{\Gamma}_{k,i}^{j,\mathcal{Z}\mathcal{Z}} \bar{u}_k^j \right] \delta \mathcal{Z}_k^i \\ + \delta u_k^\top \sum_{i=1}^n \bar{\mathcal{G}}_{k,i}^{\mathcal{Z},\top} \delta \mathcal{Z}_k^i + O(\delta \mathcal{Z}_k^3) \end{aligned} \quad (36)$$

where  $O(\delta \mathcal{Z}_k^3)$  is the third and higher order terms of  $\delta \mathcal{Z}_k$ . The notations are the same as the ones defined in Theorem 2.  $\bar{\mathcal{F}}_{k,i}^{\mathcal{Z}\mathcal{Z}}$  and  $\bar{\Gamma}_{k,i}^{j,\mathcal{Z}\mathcal{Z}}$  are the  $i$ th component of the tensor, which denotes the Hessians of the system dynamics along the optimal trajectory.

Substituting (36) into (35) and using the first-order condition for optimality of  $\delta u_k$  in the RHS of (35) give

$$\begin{aligned} \delta u_k \\ = - (R + \mathcal{B}_k^\top \bar{P}_{k+1} \mathcal{B}_k)^{-1} (R \bar{u}_k + \mathcal{B}_k^\top G_{k+1}) \\ - \underbrace{(R + \mathcal{B}_k^\top \bar{P}_{k+1} \mathcal{B}_k)^{-1} \left( \mathcal{B}_k^\top \bar{P}_{k+1} \mathcal{A}_k + \sum_{i=1}^n \bar{\mathcal{G}}_{k,i}^{\mathcal{Z},\top} \bar{G}_{k+1}^i \right)}_{K_k} \delta \mathcal{Z}_k \\ + O(\delta \mathcal{Z}_k^2) \end{aligned} \quad (37)$$

$(R \bar{u}_k + \mathcal{B}_k^\top G_{k+1}) = 0$  since the optimal  $\bar{u}_k$  satisfies (9).

Substituting (37) in (35) and equating the LHS and RHS terms based on the degree of  $\delta \mathcal{Z}_k$  yield

$$\begin{aligned} \bar{J}_k &= l(\bar{\mathcal{Z}}_k) + \frac{1}{2} \bar{u}_k^\top R \bar{u}_k + \bar{J}_{k+1} \\ \bar{G}_k &= \bar{l}_k^{\mathcal{Z}} + \mathcal{A}_k^\top \bar{G}_{k+1} \\ \bar{P}_k &= \mathcal{A}_k^\top \bar{P}_{k+1} \mathcal{A}_k + \bar{l}_k^{\mathcal{Z}\mathcal{Z}} + \sum_{i=1}^n \left[ \bar{\mathcal{F}}_{k,i}^{\mathcal{Z}\mathcal{Z}} + \sum_{j=1}^{n_u} \bar{\Gamma}_{k,i}^{j,\mathcal{Z}\mathcal{Z}} \bar{u}_k^j \right] \bar{G}_{k+1}^i \\ &\quad - K_k^\top (R + \mathcal{B}_k^\top \bar{P}_{k+1} \mathcal{B}_k) K_k. \end{aligned}$$

We have shown only terms up to the second order of  $\delta \mathcal{Z}_k$ , and higher order terms can be obtained similarly. Equation (37) gives the optimal perturbation feedback.  $\square$

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